Quantum and Gravitational Structure from a Common Event-Order Geometry

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Abstract

The VFD programme has developed two independent structural tracks from the same geometric substrate: a *quantum track* (Papers XV–XXII) in which Schrödinger-type evolution emerges as the equilibrium tangent limit of nonlinear closure-paired dynamics on the 600-cell, and a *gravitational track* (Papers XXIII–XXVII) in which temporal ordering, observer kinematics, metric separation, curvature, and source-driven field dynamics emerge from the event-order structure of the dual 600-cell system.

The present paper proposes that these two tracks admit a common interpretation as projections of a single underlying event-order geometry. Both tracks share the same foundational objects: admissible overlap events on the dual 600-cell, the closure functional $\mathcal{F}$, and the transition graph $\mathcal{G}$. The quantum track is associated with regimes in which multiple admissible configurations coexist, while the gravitational track is associated with regimes in which event ordering and accessibility geometry become dominant.

A structural correspondence table proposes parallel roles across both tracks: path integrals with geodesics, the closure functional with the source field, the Witten Hamiltonian with the graph Laplacian, and norm conservation with source balance. A regime-transition principle is formulated: a natural candidate for the quantum-to-gravitational crossover is the passage from multiplicity of admissible configurations to constraint-dominated ordering.

The paper does not derive the Einstein equations from quantum mechanics or vice versa. It proposes a structural bridge: both tracks can be understood as emerging from the same geometry, viewed under different organisational constraints.

1 Introduction

Two apparently distinct physical regimes — quantum behaviour and gravitational structure — have resisted unification within the standard framework for nearly a century. This paper does not resolve that problem. It shows something more specific: that within the VFD closure programme, the mathematical structures underlying quantum dynamics and gravitational geometry *share a common origin in the same event-order geometry.*

The two tracks

The VFD programme has constructed two tracks independently:

1. **Quantum track** (Papers XV–XXII). Starting from the stochastic closure dynamics of Paper XIV [Smart, 2026a], this track constructs phase-coherent dynamics [Smart, 2026c], proves a dissipative obstruction [Smart, 2026d], resolves it via Nelson pairing [Smart, 2026e], classifies the nonlinear residual [Smart, 2026b], synthesises the result [Smart, 2026f], and connects it to the 600-cell spectral structure [Smart, 2026g]. The central result: the Schrödinger equation emerges as the equilibrium tangent limit of nonlinear closure-paired dynamics.

2. **Gravitational track** (Papers XXIII–XXVII). Starting from the dual 600-cell phase-overlap structure identified in Paper XXII, this track derives emergent time [Smart, 2026h], observer kinematics [Smart, 2026i], metric separation [Smart, 2026j], curvature [Smart, 2026k], and source-driven field dynamics [Smart, 2026l]. The central result: a complete scaffold for gravitational structure from event-order geometry, without assuming spacetime.

Central observation

Both tracks were constructed from the same geometric substrate. This paper makes that observation explicit and asks: is this an accident of presentation, or a structural necessity?

Scope

This paper maps the shared structures, identifies the regime boundary, and formulates a unification principle. It does not derive the Einstein equations from quantum mechanics, does not produce a theory of quantum gravity, and does not claim to resolve the measurement problem. It establishes a structural bridge between two tracks that were built independently from the same geometry.

2 The Shared Substrate

Both tracks can be expressed on the same underlying substrate, comprising three objects defined in the preceding papers.

2.1 The dual 600-cell system

Paper XXII [Smart, 2026g] established that the E_8 root system contains two conjugate copies of the 600-cell (240 roots $\to$ 120 vertex pairs, 2:1 double cover via the Coxeter projection with exponents $(1, 7)$). The quantum track uses the *spectral structure* of the 600-cell graph (eigenvalues, representations, closure functional). The gravitational track uses the *overlap structure* of the dual system (admissible events from relative phase rotation).

Remark 1 (Same geometry, different access)**.** *The dual 600-cell is not two separate objects. It is one geometric system. The two tracks access different aspects of this system: spectral data (quantum) and event-order data (gravitational).*

2.2 The closure functional

The H_4-invariant closure functional

$$\mathcal{F}[\Phi] = \sum_i \log\Big(1 + e^{\beta(\Phi_i - \mu)}\Big) \tag{1}$$

was constructed explicitly in Paper XXII on the 120-vertex 600-cell graph. In the quantum track, $\mathcal{F}$ drives the stochastic dynamics: the stationary distribution $P \propto e^{-2\mathcal{F}/\sigma^2}$ (Paper XIV [Smart, 2026a]), the Nelson pairing pairs forward/backward processes under $\mathcal{F}$-gradient drift (Paper XVIII [Smart, 2026e]), and the Witten Hamiltonian $H_{\mathrm{W}} = -(\sigma^2/2)\Delta + V_{\mathrm{W}}$ with $V_{\mathrm{W}} = \frac{1}{2}|\nabla\mathcal{F}|^2 - \frac{\sigma^2}{2}\Delta\mathcal{F}$ governs the equilibrium tangent dynamics.

In the gravitational track, $\mathcal{F}$ determines which vertex configurations are admissible: an overlap event is admissible when the local closure residual falls below threshold. The source field $S(e)$ of Paper XXVII [Smart, 2026l] measures deviation from uniform closure, and the field equation $\Delta_{\mathcal{G}} u = S - \bar{S}$ describes how this deviation organises the accessibility geometry.

Remark 2. *The same functional $\mathcal{F}$ appears in both tracks. In the quantum track it is the potential energy of the stochastic dynamics. In the gravitational track it determines which events exist and how they are distributed. This strongly suggests that both tracks are projections of the same geometric system.*

2.3 The transition graph

The event set $\mathcal{E}$ carries a natural graph structure $\mathcal{G} = (\mathcal{E}, E_H)$ where edges connect events that are related by admissible transitions (Paper XXIII [Smart, 2026h]). In the quantum track, the graph Laplacian $\Delta_{\mathcal{G}}$ appears in the Witten Hamiltonian (as the kinetic operator). In the gravitational track, the same $\Delta_{\mathcal{G}}$ appears in the field equation (as the operator relating source to potential).

Definition 3 (Shared substrate)**.** *The shared substrate of the VFD programme is the triple $(\mathcal{E}, \mathcal{F}, \mathcal{G})$: the event set, the closure functional, and the transition graph, all derived from the dual 600-cell system.*

3 The Quantum Regime

The quantum track is characterised by the following structural features, all established in the preceding papers.

3.1 Multiplicity of admissible configurations

When the relative phase parameter θ admits multiple overlapping configurations simultaneously, the system exhibits *superposition*: the closure functional has multiple comparable minima, and the stochastic dynamics explores all of them. This is the regime in which Paper XV's interference [Smart, 2026c] and Paper XVIII's Nelson pairing [Smart, 2026e] operate.

3.2 Phase structure and interference

In this regime, the complexified Onsager–Machlup action of Paper XV assigns phase-rotation frequency $\Omega = \mathcal{F}/\sigma^2$ and amplitude $A = e^{-\mathcal{F}/\sigma^2}$ to each path. Paths between the same endpoints carry different phases and interfere. This is the structural origin of quantum-type behaviour within the closure framework.

3.3 Unitary-type evolution

At the Paper XIV equilibrium, the paired forward/backward process yields the Schrödinger equation with Witten Hamiltonian H_{W} (Paper XVIII). The evolution is norm-conserving (Paper XIX [Smart, 2026b]). Away from equilibrium, the dynamics is nonlinear, with a modulus-curvature residual δU_{rel} that is gauge-invariant and perturbatively small near equilibrium.

3.4 Summary

> The quantum regime is characterised by: multiplicity of admissible configurations, path-level phase structure producing interference, and norm-conserving evolution governed by the closure functional through the Witten Hamiltonian.

4 The Gravitational Regime

The gravitational track is characterised by the following structural features, all established in Papers XXIII–XXVII.

4.1 Event ordering

When the birth-angle parameter θ induces a strict partial order on admissible events (Paper XXIII [Smart, 2026h]), the system acquires a temporal structure. Events are no longer simultaneously accessible; they are ordered by when they become admissible. This is the regime in which path multiplicity is resolved into sequential accessibility.

4.2 Observer frames and kinematics

Within the ordered event set, different linear extensions define different observer frames (Paper XXIV [Smart, 2026i]). Simultaneity is frame-dependent. Time dilation is exhibited. These are kinematic consequences of the ordering, not postulates.

4.3 Metric, curvature, and dynamics

The transition graph $\mathcal{G}$ carries a natural distance structure (Paper XXV [Smart, 2026j]), curvature indicators from non-uniform accessibility (Paper XXVI [Smart, 2026k]), and a source-driven field equation $\Delta_{\mathcal{G}} u = S - \bar{S}$ (Paper XXVII [Smart, 2026l]). The complete chain is: source $\to$ potential $\to$ curvature $\to$ geodesic concentration.

4.4 Summary

The gravitational regime is characterised by: constraint-driven event ordering, frame-dependent kinematics, metric separation, curvature from non-uniform accessibility, and source-driven geometric response governed by the graph Laplacian field equation.

5 Structural Correspondence

The following table records the principal proposed correspondences suggested by the preceding constructions. Each row identifies objects that arise from the same geometric substrate but serve different functions in their respective regimes.

Table 1: Proposed structural correspondence between quantum and gravitational tracks.

Shared object	Quantum role	Gravitational role
Event set $\mathcal{E}$	Configuration space	Causal event set
Closure functional $\mathcal{F}$	Stochastic potential	Admissibility criterion / source
Graph Laplacian $\Delta_{\mathcal{G}}$	Kinetic operator in H_{W}	Field-equation operator
Paths on $\mathcal{G}$	Interference (sum over paths)	Geodesics (shortest paths)
$\mathcal{F}$-gradient	Drift in Nelson pairing	Source field $S(e)$
Norm conservation	$\lVert\Psi\rVert^2 = \text{const}$	$\sum(S - \bar{S}) = 0$
Equilibrium	Stationary distribution R^2_{st}	Vacuum (uniform S, flat u)
Non-equilibrium	Nonlinear residual δU_{rel}	Curvature $K_{\mathrm{pot}} > 0$

Remark 4 (Status of the correspondence)**.** *Table 1 is a structural map, not a derivation. It identifies objects that arise from the same substrate and play parallel roles. The correspondence is exhibited, not proved to be necessary. Whether it reflects a deeper mathematical duality is an open question.*

5.1 Path structure: interference vs geodesics

This correspondence deserves elaboration because it connects the most characteristic features of each regime.

In the quantum regime, the system explores all paths on $\mathcal{G}$ between two events. Each path carries a phase $e^{i\Omega\ell}$ where ℓ is the path cost and $\Omega = \mathcal{F}/\sigma^2$. The total amplitude is a sum over paths, producing interference.

In the gravitational regime, the system has resolved into an ordered configuration. Propagation follows geodesics — shortest paths on $\mathcal{G}$ weighted by the accessibility potential $u(e)$. Geodesics concentrate toward regions of high potential gradient (Paper XXVII).

Quantum: all paths contribute, weighted by phase.
Gravitational: dominant path selected by geometric cost.

A natural interpretation is that, as constraint ordering strengthens, the effective path contribution becomes increasingly concentrated near geodesic structure. If correct, this would be the geometric analogue of the stationary-phase (WKB) limit in standard quantum mechanics.

5.2 Operator structure: Witten Hamiltonian vs graph Laplacian

In the quantum regime, the dynamics at equilibrium is governed by the Witten Hamiltonian:

$$H_{\mathrm{W}} = -\frac{\sigma^2}{2}\Delta_{\mathcal{G}} + V_{\mathrm{W}}, \qquad V_{\mathrm{W}} = \frac{1}{2}|\nabla\mathcal{F}|^2 - \frac{\sigma^2}{2}\Delta\mathcal{F} \tag{2}$$

In the gravitational regime, the field equation is:

$$\Delta_{\mathcal{G}}\, u(e) = S(e) - \bar{S} \tag{3}$$

Both are governed by the same operator $\Delta_{\mathcal{G}}$ on the same graph. The Witten Hamiltonian adds the closure potential V_{W}, which encodes the local curvature of the closure landscape. The field equation uses $\Delta_{\mathcal{G}}$ directly, with the source S playing the role that V_{W} plays in the quantum case: driving the system away from uniformity.

Remark 5. *The graph Laplacian $\Delta_{\mathcal{G}}$ is the shared kinetic/operator core in both tracks. In the quantum regime it provides the kinetic term in H_{W}, which generates evolution. In the gravitational regime it mediates geometric response through the field equation. The same operator, on the same graph, in two roles.*

6 Regime Transition

The two regimes are not separated by a discontinuity. They are connected by a continuous parameter: the *degree of constraint ordering* in the event set.

6.1 The ordering parameter

Definition 6 (Constraint ordering strength)**.** *For an event set $\mathcal{E}$ with partial order $\preceq$, define the ordering fraction:*

$$\eta = \frac{|\{(e_1, e_2) : e_1 \prec e_2\}|}{\binom{|\mathcal{E}|}{2}} \tag{4}$$

the fraction of event pairs that are comparable under $\preceq$.

- $\eta = 0$: no ordering (antichain). All events are simultaneously admissible. Maximum path multiplicity. Pure quantum regime.
- $\eta = 1$: total ordering (chain). All events are sequentially ordered. Single dominant path. Pure gravitational regime.
- $0 < \eta < 1$: partial ordering. Both quantum and gravitational features coexist.

6.2 Weak-ordering regime: quantum

In the weak-ordering regime (few events, high symmetry), the event set is weakly ordered. Many events are simultaneously admissible. The path sum is dominated by interference, and the Nelson-paired dynamics yields Schrödinger-type evolution. This is the regime of Papers XV–XXI.

6.3 Strong-ordering regime: gravitational

In the strong-ordering regime (many events, constraint accumulation), the event ordering becomes strong. Most event pairs are comparable. Path multiplicity is suppressed, geodesics dominate, and the accessibility geometry acquires metric, curvature, and field-equation structure. This is the regime of Papers XXIII–XXVII.

6.4 The crossover

Proposition 7 (Combinatorial regime indicator)**.** *For a finite event set $\mathcal{E}$ with partial order $\preceq$ and ordering fraction η:*

1. *If η is small, the number of linear extensions of $\preceq$ is large, and the system admits many distinct observer frames and path configurations (quantum multiplicity).*
2. *If η is close to 1, the number of linear extensions is small (approaching 1 for a total order), and the system admits essentially one frame and one dominant path class (gravitational concentration).*

Proof. The number of linear extensions of a partial order is bounded above by $n!/\prod c_i$ where c_i are chain lengths, and is maximised for the antichain ($n!$ extensions) and minimised for the total order (1 extension). Both bounds are standard in order theory. The correspondence between extension count and observer multiplicity follows from the fact that each linear extension defines a distinct observer frame (Paper XXIV). This supports the interpretation of large extension count as a proxy for path and frame multiplicity. □

Remark 8 (Scale and ordering)**.** *The claim is not that "quantum = small" and "gravity = large" in any absolute sense. The claim is structural: as constraint ordering strengthens (by whatever mechanism), the system transitions from the regime in which quantum-type behaviour dominates to the regime in which gravitational-type structure dominates. The ordering fraction η parameterises this transition. Whether η correlates with physical scale is an empirical question that the framework does not answer.*

7 The Unification Principle

The preceding sections establish:

1. Both tracks arise from the same substrate $(\mathcal{E}, \mathcal{F}, \mathcal{G})$.
2. The same objects (closure functional, graph Laplacian, paths) play parallel roles in both tracks.
3. The ordering fraction η provides a candidate parameter for the regime boundary.

This motivates the following structural principle:

> **Unification principle.** Quantum behaviour and gravitational structure are not separate physical domains. They are two organisational regimes of the same event-order geometry:
>
> - *Quantum*: multiplicity of admissible configurations, path interference, norm-conserving evolution.

- *Gravitational*: constraint-driven event ordering, geodesic concentration, source-driven geometric response.

A natural candidate parameter for the transition between them is the degree of constraint ordering in the event set.

Remark 9 (Epistemic status)**.** *This is a structural principle, not a theorem. It is motivated by the explicit constructions of Papers XV–XXVII and the correspondence table of Section 5. It does not constitute a proof of unification, a derivation of quantum gravity, or a replacement for either quantum mechanics or general relativity. It identifies a structural pattern within the VFD programme and proposes it as a candidate organising principle.*

8 Crystallisation as the Bridge

The crystallisation operator — the founding object of the VFD programme — acquires a new interpretation in this context.

8.1 The crystallisation functional

The crystallisation operator $C[\Psi] = \operatorname{argmin}_\Phi \mathcal{F}[\Phi]$ maps an unresolved configuration to the stable attractor of the closure functional. In the original formulation, this was a model of quantum state selection: the system evolves under gradient flow of $\mathcal{F}$ until it reaches a constraint-compatible configuration.

8.2 Reinterpretation

In the unified picture, crystallisation is not merely a selection mechanism. It is the process by which the event-order geometry transitions from the quantum regime to the gravitational regime:

- **Before crystallisation**: multiple admissible configurations coexist. The event set is weakly ordered. Quantum-type path interference dominates.
- **During crystallisation**: the closure functional drives convergence. Constraint ordering strengthens. The ordering fraction η increases.
- **After crystallisation**: the system has settled into a strongly ordered configuration. Geodesics dominate. Metric, curvature, and field-equation structure emerge.

  Crystallisation provides the geometric bridge by which quantum multiplicity can be interpreted as resolving into gravitational order.

Remark 10. *This interpretation is structural, not dynamical in the sense of specifying a time-dependent process. The "before/during/after" language describes the logical relationship between regimes, not necessarily a temporal sequence. Whether crystallisation occurs in time, or whether time itself emerges from the crystallisation process (as Paper XXIII suggests), is a question the framework poses but does not resolve.*

9 Discussion and Limitations

9.1 What is established

- Both the quantum and gravitational tracks of the VFD programme arise from the same geometric substrate: the dual 600-cell event-order system $(\mathcal{E}, \mathcal{F}, \mathcal{G})$.
- A structural correspondence maps objects across both tracks (Table 1).
- The regime boundary is parameterised by the ordering fraction η (Definition 6).
- The crystallisation operator provides a structural bridge between regimes.
- The graph Laplacian $\Delta_{\mathcal{G}}$ serves as the shared kinetic/operator core in both tracks.
- Path interference (quantum) and geodesic concentration (gravitational) are structural limits of the same path structure on $\mathcal{G}$.

9.2 What is not established

- No derivation of the Einstein equations from quantum mechanics or vice versa.
- No theory of quantum gravity. The paper identifies a structural bridge, not a dynamical unification.
- No continuum limit. Whether the discrete structures converge to continuous quantum field theory or general relativity is open.
- No experimental predictions from the unification principle itself (though both constituent tracks make independent structural predictions).
- No proof that the correspondence in Table 1 is unique or necessary. Alternative mappings may exist.
- The ordering parameter η is defined on finite event sets. Its behaviour in any continuum or thermodynamic limit is not analysed.
- The regime characterisation (Proposition 7) is qualitative. No sharp phase-transition threshold is identified.

9.3 Comparison with existing approaches

Table 2: Comparison with other unification approaches.

Approach	Starting point	Unification mechanism	Status
String theory	Continuous strings	Extra dimensions, dualities	No unique vacuum
Loop quantum gravity	Spin networks	Discrete spacetime	Classical limit open
Causal set theory	Discrete partial orders	Order $\to$ geometry	QM recovery open
VFD (this paper)	Closure on 600-cell	Regime transition on $\mathcal{G}$	Structural bridge

Remark 11. *The VFD approach is closest in spirit to causal set theory, which also begins from discrete partial orders. The key structural difference is that VFD derives the partial order from the 600-cell overlap geometry (Paper XXIII), whereas causal set theory takes the partial order as fundamental. Whether this difference is physically significant is an open question.*

10 Programme Summary

The VFD closure programme now comprises the following completed tracks:

Table 3: Complete programme structure.

Track	Papers	Core result	Regime
Crystallisation	Flagship + A–D	Deterministic state selection	Selection
Mass / constraint	I–XIII	Particle masses from 600-cell eigenvalues	Spectral
Quantum recovery	XV–XXI	Schrödinger as equilibrium tangent limit	Quantum
Spectral bridge	XXII	Mass, gauge, constants from $\mathcal{F}$	Spectral–quantu
GR scaffold	XXIII–XXVII	Time, metric, curvature, dynamics	Gravitational
Unification	XXVIII (this paper)	Structural bridge across regimes	Both

11 Future Directions

The structural bridge identified here opens several directions for future work:

1. **Continuum limit.** Does the discrete framework converge to known continuum theories (quantum field theory, general relativity) in appropriate limits? This is the defining technical challenge.

2. **Regime-transition dynamics.** The ordering fraction η parameterises the crossover, but the dynamics of the transition — how constraint ordering strengthens — is not specified. A dynamical theory of the quantum-to-gravitational crossover would substantially strengthen the framework.

3. **Coupling constant.** The gravitational field equation $\Delta_{\mathcal{G}} u = S - \bar{S}$ has no coupling constant (proportionality set to 1). Identifying a Newton-constant analogue from the 600-cell spectral data is an open target.

4. **Observable signatures.** Both tracks make structural predictions independently. Whether the unification principle itself generates new, testable predictions — beyond those of the constituent tracks — is the key question for experimental relevance.

12 Conclusion

The VFD closure programme has constructed quantum and gravitational structure from the same geometric substrate: the dual 600-cell event-order system. The quantum track (Papers XV–XXII) derives Schrödinger-type evolution from closure-paired dynamics. The

gravitational track (Papers XXIII–XXVII) derives temporal ordering, metric separation, curvature, and field dynamics from event-order geometry.

This paper has shown that these constructions admit a common structural interpretation. Both tracks share the same foundational triple $(\mathcal{E}, \mathcal{F}, \mathcal{G})$. The same core objects — the closure functional, the graph Laplacian, and the path structure — admit parallel interpretations in both regimes. A natural candidate for the transition between quantum and gravitational behaviour is the strengthening of constraint ordering in the event set, parameterised by the ordering fraction η.

The result is a structural bridge, not a theory of quantum gravity. Whether the bridge can be made dynamical, whether it converges to known physics in appropriate limits, and whether it generates new experimental predictions are the defining open questions for the next stage of the programme.

What can be stated is this: within the VFD framework, the distinction between quantum and gravitational structure is not fundamental. It is a consequence of how event accessibility is organised under constraint.

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